Ericsson PE Node BTM-PE-05 Standard Operating Procedure for Hardware/Environmental Fault (ASIC Memory Parity Errors, Line Card Crashes, Power Supply Redundancy Loss) Alarm ALARM_3556

Node: BTM-PE-05 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_3556

Description

The Ericsson PE node BTM-PE-05, operating within the BTM Layout Aggregation site, is susceptible to hardware/environmental faults leading to ASIC memory parity errors, line card crashes, and power supply redundancy loss. This is primarily due to the intricacies of the node's hardware architecture, which relies heavily on the Advanced ASIC (Application-Specific Integrated Circuit) to manage traffic and control plane operations. The ASIC's memory parity errors can propagate and cause a cascading effect, ultimately resulting in a line card crash. Furthermore, the loss of power supply redundancy can exacerbate the situation, hindering the node's ability to recover. The Ericsson CLI dialect is utilized in this SOP to navigate and resolve the alarm ALARM_3556, ensuring that the node is restored to a stable state.

Pre-requisite Checks

- Check: Verify that remote hands are dispatched to the BTM Layout Aggregation site to facilitate physical access and troubleshooting.

- Check: Confirm that the redundant link status is intact and functioning correctly.

- Check: Ensure that the Ericsson CLI is accessible and configured for the BTM-PE-05 node.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system redundancy, Expected Output: Redundancy status: ACTIVE, Description: This command verifies the redundancy status of the power supply, allowing us to determine if the issue is related to power supply redundancy loss.

Command: show system asic status, Expected Output: ASIC status: ONLINE, Description: This command checks the status of the ASIC, which is critical for managing traffic and control plane operations. A
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faulty ASIC can cause ASIC memory parity errors and line card crashes.

Command: show system linecard status, Expected Output: Linecard
status: ONLINE, Description: This command verifies the status of the
linecard, which is responsible for transmitting and receiving data
packets. A faulty linecard can cause line card crashes.
```

Resolution Steps

Step 1: Step: Power Cycle the Node, Command: system power cycle, Description: Performing a power cycle will restart the node and potentially resolve the issue. However, this step should be performed with caution and only after verifying that the node is not in a critical state.

Step 2: Step: Swap the Power Supply, Command: system redundancy power supply swap, Description: Swapping the power supply will ensure that the node has a functioning power supply redundancy, reducing the risk of power supply redundancy loss.

Step 3: Step: Replace the Faulty ASIC, Command: system asic replace, Description: Replacing the faulty ASIC will resolve the ASIC memory parity errors and prevent further line card crashes.

Step 4: Step: Replace the Faulty Linecard, Command: system linecard replace, Description: Replacing the faulty linecard will resolve the line card crashes and ensure that data packets are transmitted and received correctly.

Post-Check Verification

```
Command: show system status, Expected Output: System status: NORMAL,
Description: This command verifies that the node is functioning
normally and that the alarm ALARM_3556 has been cleared.

Command: show system redundancy, Expected Output: Redundancy status:
ACTIVE, Description: This command verifies that the power supply
redundancy is intact and functioning correctly.

Command: show system asic status, Expected Output: ASIC status:
ONLINE, Description: This command verifies that the ASIC is
functioning correctly and that ASIC memory parity errors have been
resolved.

Command: show system linecard status, Expected Output: Linecard
status: ONLINE, Description: This command verifies that the linecard
is functioning correctly and that line card crashes have been
resolved.
```

Roll-back Procedures

- Step: Restore the Original Configuration, Command: system configuration restore, Description: Restoring the original configuration will ensure that the node returns to its previous state and that any changes made during the troubleshooting process are reverted.

- Step: Reboot the Node, Command: system reboot, Description: Rebooting the node will ensure that any changes made during the troubleshooting process are applied correctly and that the node functions as expected.